PRACTICAL NO. 4

Aim: To import data warehouse data from SQL Server into Microsoft Excel and create Pivot Table and Pivot Chart for analysis.

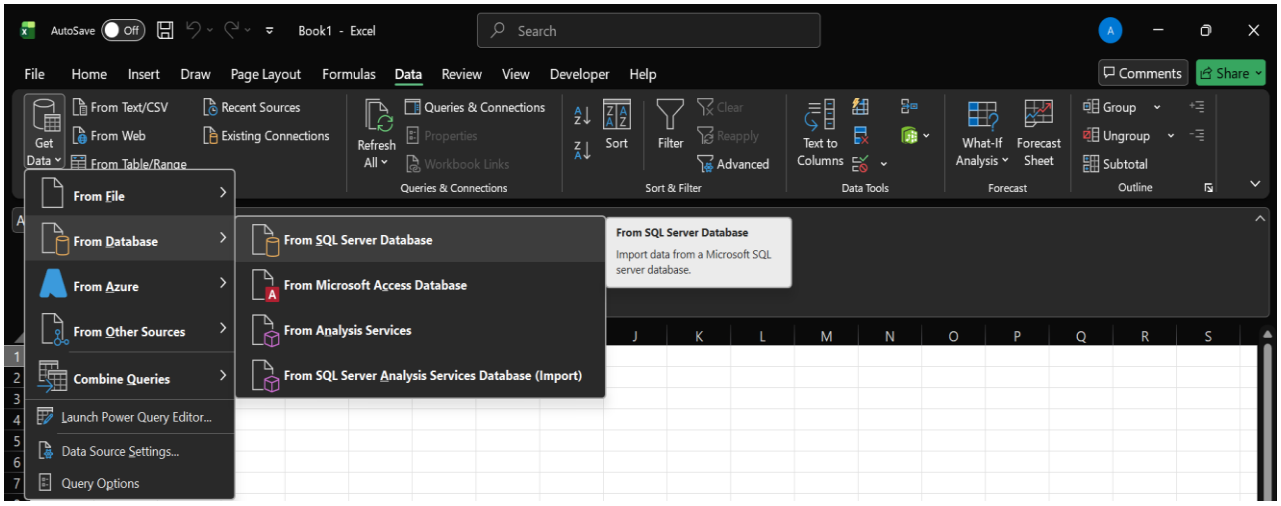
Tools Used:

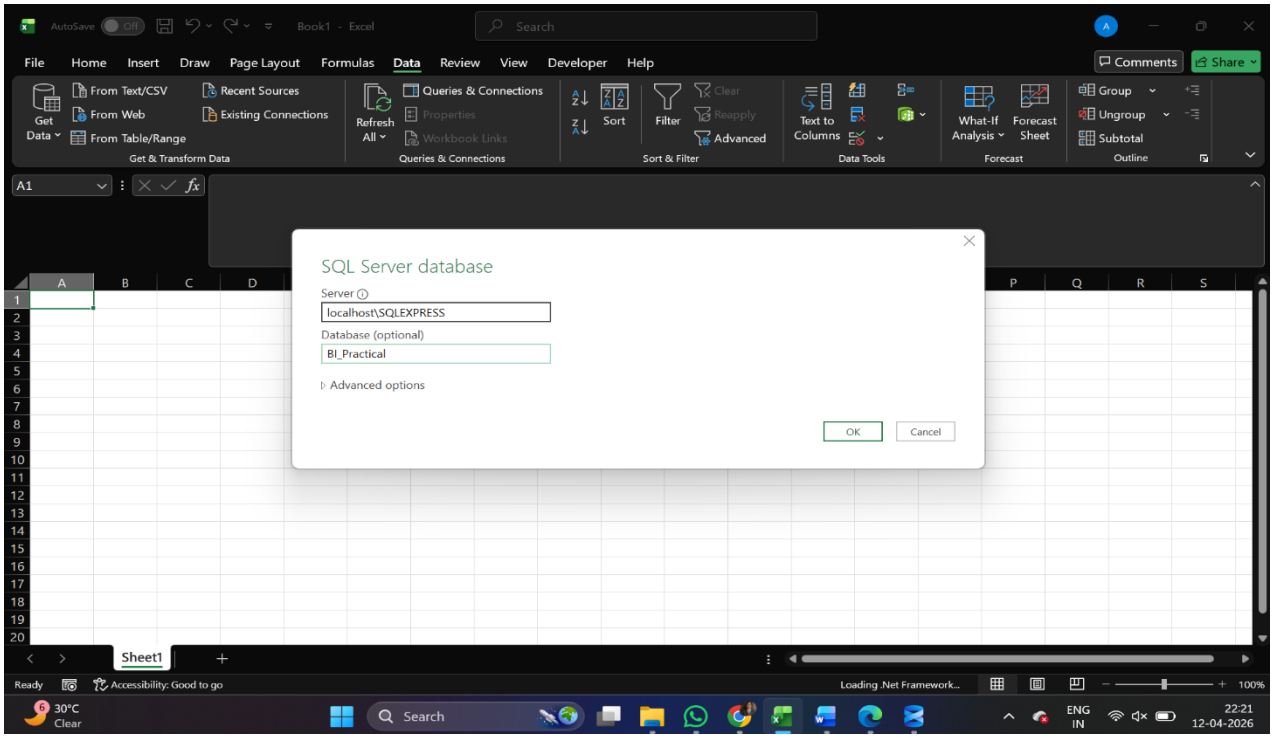
- **Microsoft SQL Server**
- **SQL Server Management Studio (SSMS)**
- **Microsoft Excel**

Procedure:

Step 1: Connect Excel to SQL Server

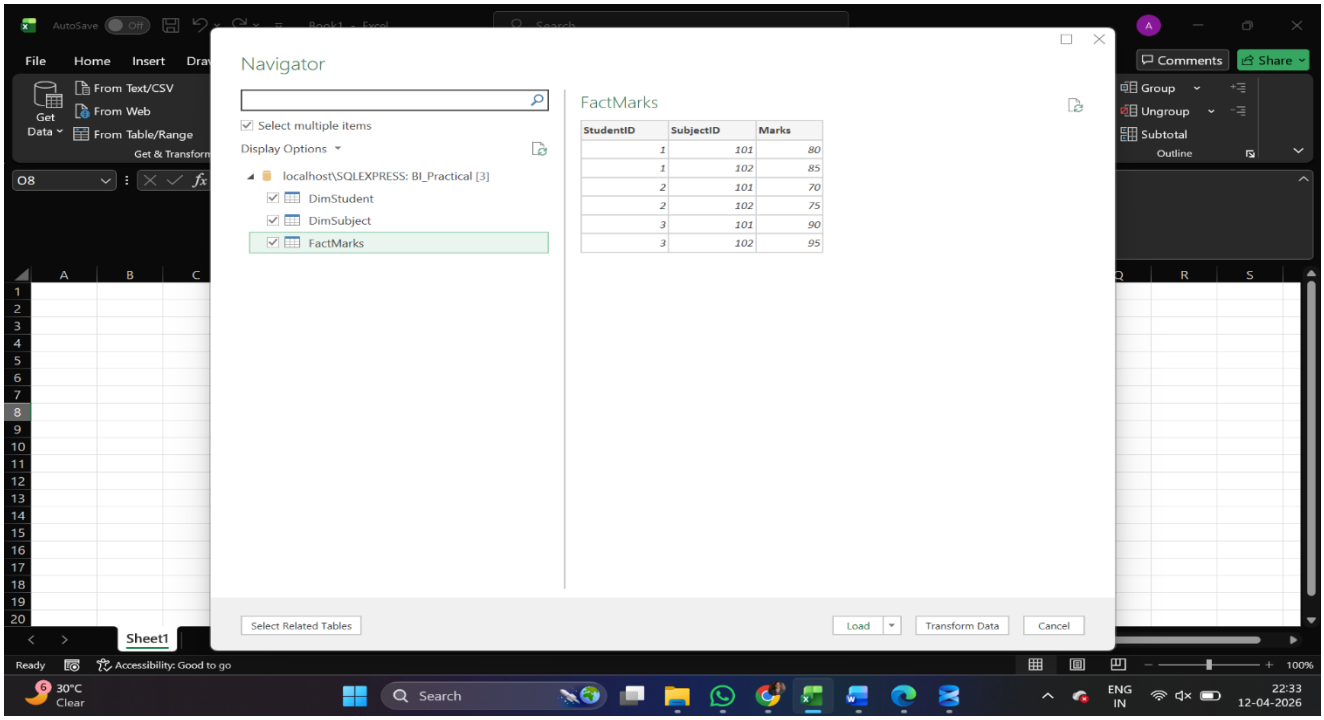
- **Open Excel**
- **Go to Data → Get Data → From SQL Server Database**
- **Enter server name: localhost\SQLEXPRESS**
- **Select database: BI_Practical**





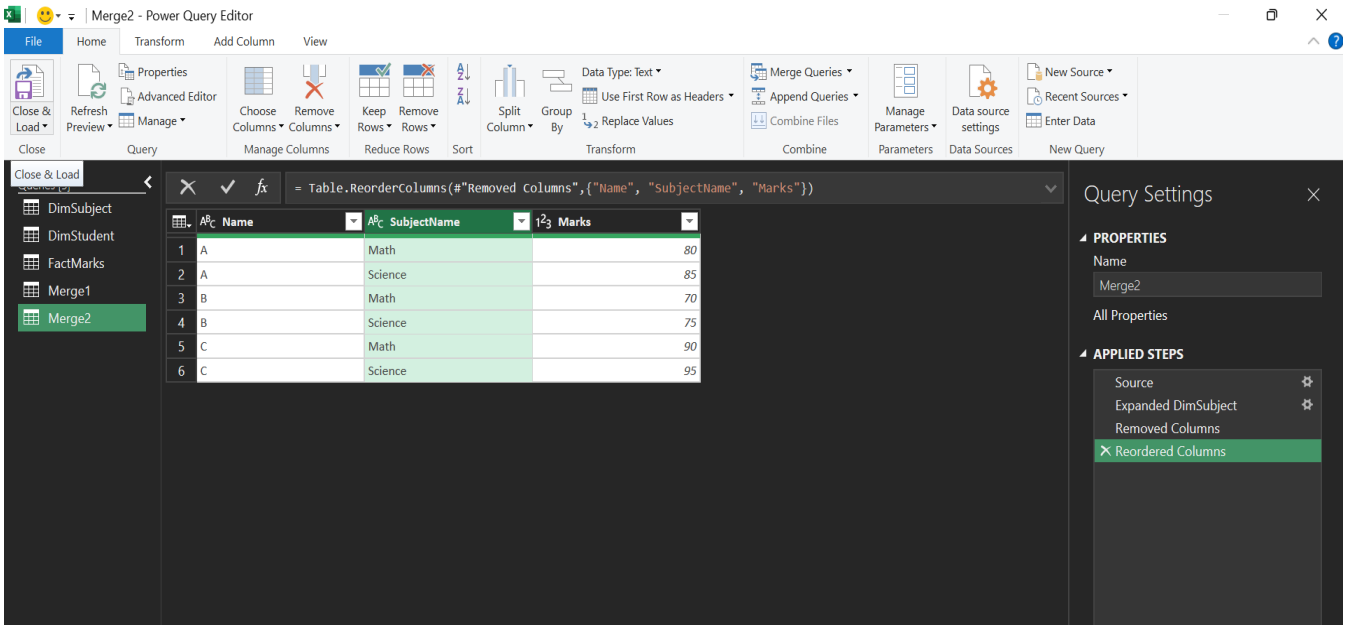
Step 2: Select Tables

- **In Navigator, select:**
  - **DimStudent**
  - **DimSubject**
  - **FactMarks**
- **Click Transform Data**



Step 3: Merge Tables

- **Merge FactMarks with DimStudent using StudentID**
- **Expand and select Name**
- **Merge with DimSubject using SubjectID**
- **Expand and select SubjectName**
- **Final table contains: Name, SubjectName, Marks**



Step 4: Load Data into Excel

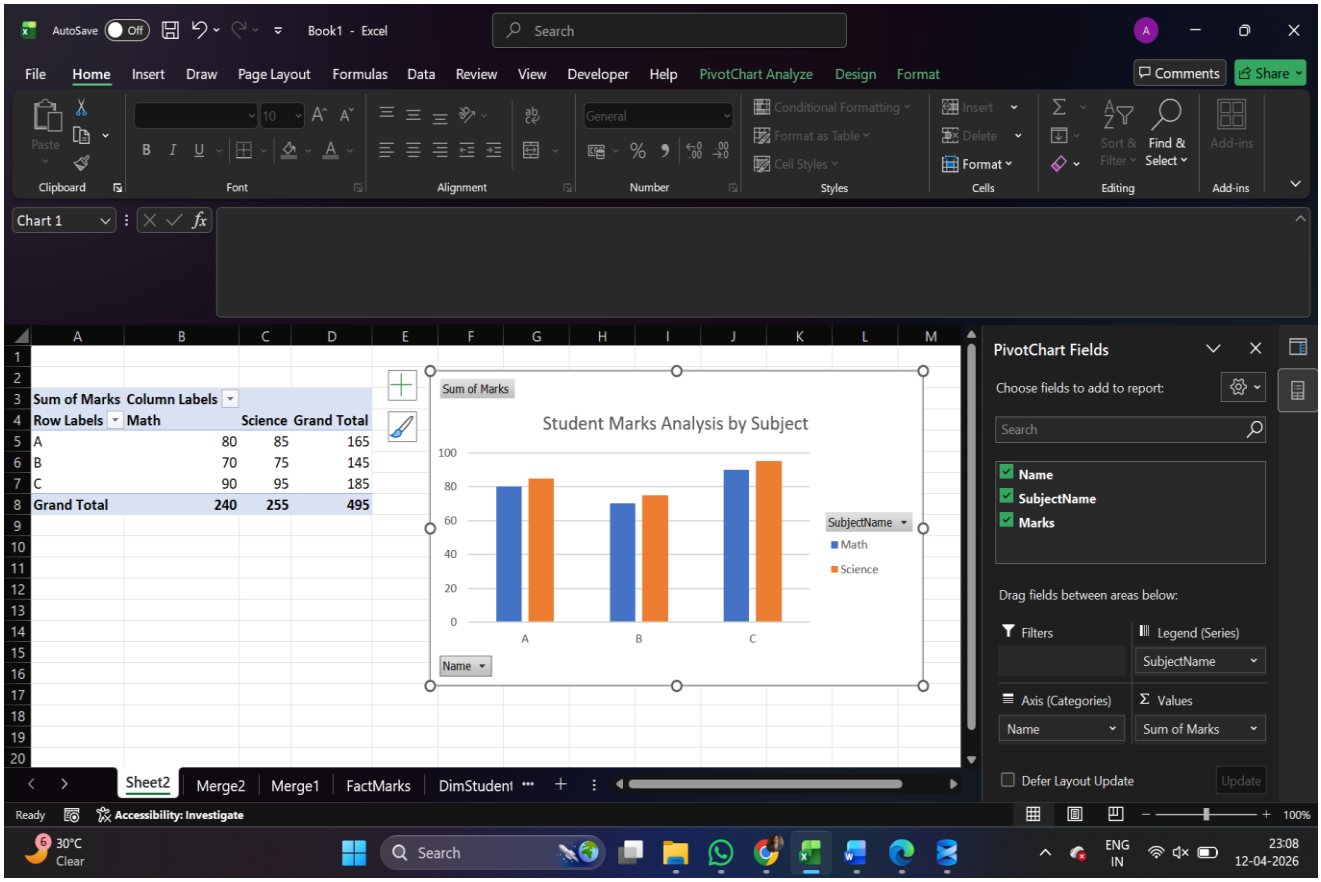
- **Click Close & Load**
- **Data is loaded into Excel sheet**

Step 5: Create Pivot Table

- **Select the table**
- **Go to Insert → Pivot Table → New Worksheet**
- **Set fields:**
  - **Rows → Name**
  - **Columns → SubjectName**
  - **Values → Marks (Sum)**

Step 6: Create Pivot Chart

- **Select Pivot Table**
- **Go to Insert → Chart → Column Chart**



Result:

- **Data warehouse data was successfully imported into Excel.**
- **Pivot Table was created to summarize student marks.**
- **Pivot Chart was generated to visualize the data effectively.**

Conclusion:

- **Excel can be used for analyzing data warehouse data.**
- **Pivot Tables help in summarizing data efficiently.**
- **Pivot Charts provide better visualization for decision making.**